

CSA0613-Design and Analysis of Algorithms

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Q33. Debugging Linear Search Code (Incorrect Index Variable Update)

Aim

To debug the given Linear Search program so that it returns the first occurrence of the target element immediately after it is found.

Problem in the Given Code

The code updates the variable pos whenever the key is found. If the return statement is placed outside the loop (as intended by the question), the function returns the last occurrence instead of the first occurrence.

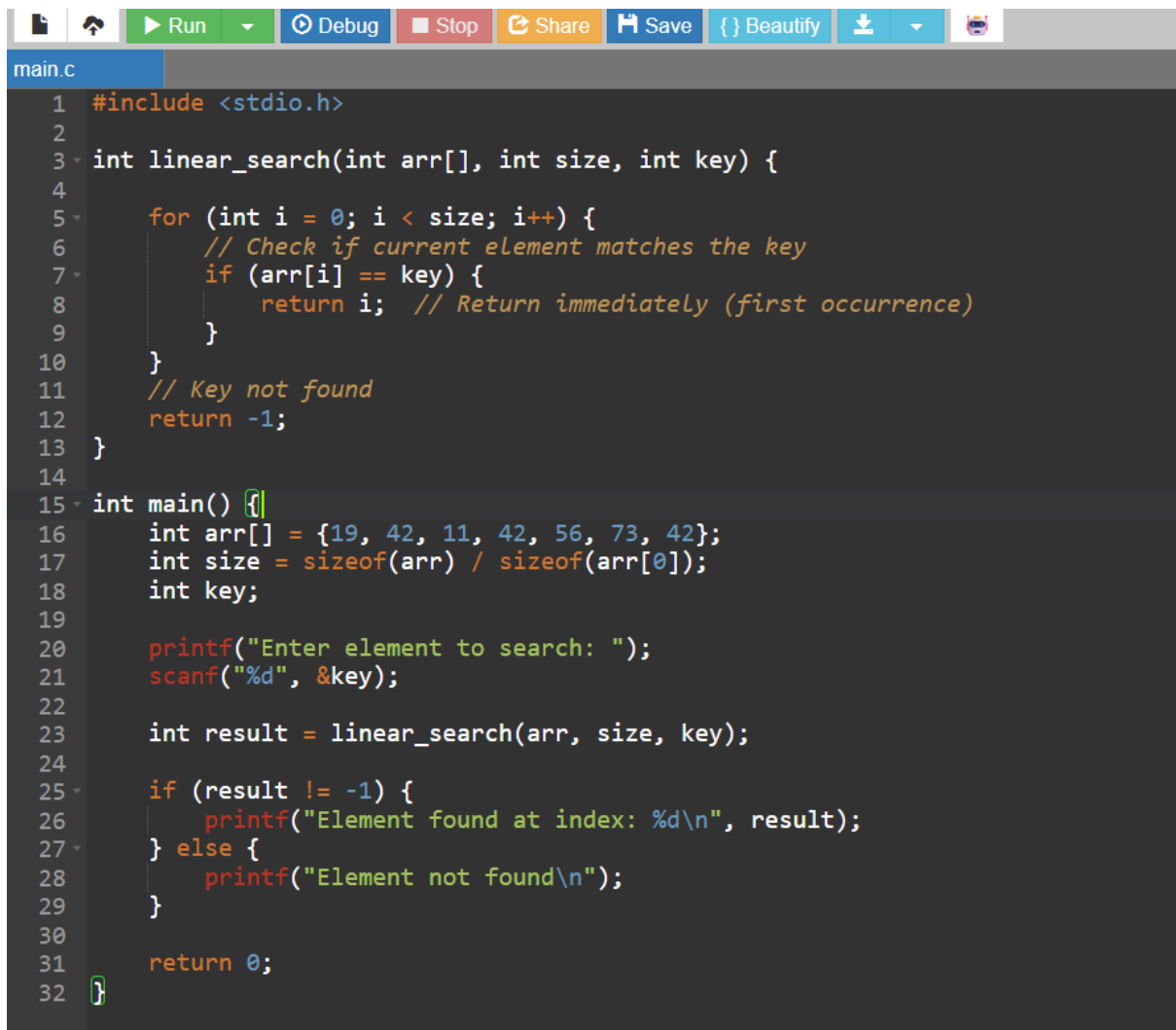
Incorrect Code:

```
def linear_search(arr, key):  
    pos = -1  
    for i in range(len(arr)):  
        if arr[i] == key:  
            pos = i  
    return pos
```

Reason:

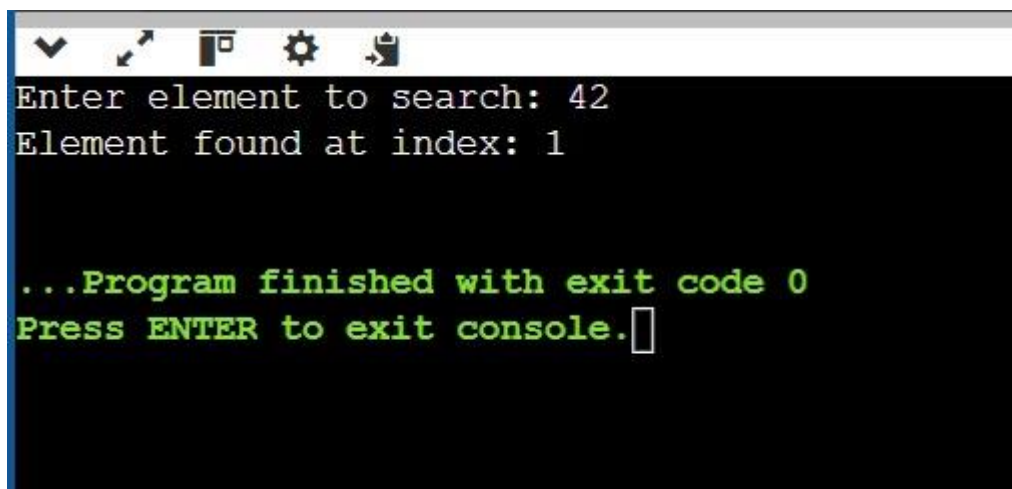
- pos keeps getting updated whenever another matching element is found.
- Hence, after the loop ends, pos contains the last matching index.

Corrected Python Code



```
1 #include <stdio.h>
2
3 int linear_search(int arr[], int size, int key) {
4     for (int i = 0; i < size; i++) {
5         // Check if current element matches the key
6         if (arr[i] == key) {
7             return i; // Return immediately (first occurrence)
8         }
9     }
10    // Key not found
11    return -1;
12 }
13
14
15 int main() {
16     int arr[] = {19, 42, 11, 42, 56, 73, 42};
17     int size = sizeof(arr) / sizeof(arr[0]);
18     int key;
19
20     printf("Enter element to search: ");
21     scanf("%d", &key);
22
23     int result = linear_search(arr, size, key);
24
25     if (result != -1) {
26         printf("Element found at index: %d\n", result);
27     } else {
28         printf("Element not found\n");
29     }
30
31     return 0;
32 }
```

Sample Output



```
Enter element to search: 42
Element found at index: 1

...Program finished with exit code 0
Press ENTER to exit console.
```

Explanation

- The list is checked one element at a time.
- As soon as the target element is found, the function immediately returns its index.
- This ensures the first occurrence is returned.
- If the element is not found after checking the entire list, -1 is returned.

Algorithm

1. Start.
2. Read the array and search key.
3. Traverse the array from index 0.
4. If the current element equals the key, return its index immediately.
5. If the loop finishes without finding the key, return -1.
6. Display the result.
7. Stop.

Complexity Analysis

- Best Case: $O(1)$ (element found at the first position)
- Average Case: $O(n)$
- Worst Case: $O(n)$ (element at the last position or not present)
- Space Complexity: $O(1)$

Conclusion

The debugging process corrected the logic by returning the index immediately after finding the first matching element. This guarantees the correct behavior of Linear Search and improves code readability and control flow.